Timothy W. Randolph

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Appointments

Harvey Mudd College, Claremont, CA — 2024-
Assistant Professor, Department of Computer Science

Education

Columbia University, New York, NY — 2018-2024
Ph.D., Computer Science — 2024
Thesis: "Exact Algorithms for Subset Sum and Subset Balancing Problems"
M.Phil., Computer Science — 2022
M.S., Computer Science — 2019
Advised by Xi Chen *and* Rocco A. Servedio

Williams College, Williamstown, MA — 2014-2018
B.A., Computer Science (Highest Honors), Mathematics (Honors), Philosophy — 2018
Thesis: "(k, p)-Planar Graphs: A Generalized Planar Representation for Cluster Graphs"
Advised by William J. Lenhart

Teaching Experience

Harvey Mudd College, Claremont, CA

Instructor for CSCI 145: Advanced Topics in Algorithms — S26, S27
New course developed for S26. "Peer-to-peer" tutorial course in algorithms and complexity; heavy emphasis on presentation, communication, collaboration, and independent learning.

Instructor for CSCI 140 / MATH 168: Algorithms — F24, S25, F25, S26, F26, S27
Comprehensive undergraduate algorithms course.

Instructor for WRIT 1: Introduction to Academic Writing — F25
Freshman course focused on developing academic writing skills.

Advisor for CSCI 183/184: Computer Science Clinic I & II — F24/S25, F25/S26, F26/S27
Year-long project-based capstone course with 4-6 students. Advised Hawai'i Keiki Museum (F24/S25), UC Berkeley (F25/S26), and LS Power (F26/S27).

Columbia University, New York, NY

Instructor for COMS 3261: Computer Science Theory — S21, S22, S23
Undergraduate automata, computability and complexity course.

Columbia CTL Teaching Development Program (Advanced Track) — 2019-2022
Columbia CTL Teaching Observation Fellow — 2019-2020
Columbia CTL Innovative Teaching Summer Institute (ITSI) — 2019

Journal and Conference Publications[†]

My two primary research interests are (1) computer science theory (exact and parameterized algorithms, algorithmic combinatorics) and (2) computer science theory education. Great

Beating Meet-in-the-Middle for Sumset Balancing Problems STOC 2026
Tim Randolph and Karol Węgrzycki
2026 Symposium on Theory of Computing
View Online

Testing Sumsets is Hard ESA 2025
Xi Chen, Shivam Nadimpalli, Tim Randolph, Rocco Servedio, and Or Zamir
2025 European Symposium on Algorithms
View Online

A Survey of Undergraduate Theory of Computing Curricula SIGCSE Virtual 2024
Ryan E. Dougherty (working group co-lead), Tim Randolph (working group co-lead), Tzu-Yi Chen, Jeff Erickson, Matthew Ferland, Dennis Komm, Jonathan Liu, Timothy Ng, Smaranda Sandu, Michael Shindler, Edward Talmage, and Thomas Zeume
2024 ACM Virtual Global Computing Education Conference
View Online

Parameterized Algorithms on Integer Sets with Small Doubling: Freiman's Theorem, Subset Sum and k-Sum ESA 2024
Tim Randolph and Karol Węgrzycki
2024 European Symposium on Algorithms
View Online

Experience Report: Participatory Governance in the CS Theory Classroom
Tim Randolph SIGCSE 2024
2024 ACM Technical Symposium on Computer Science Education
View Online

Subset Sum in $2^{n/2}/poly(n)$ Time RANDOM 2023
Xi Chen, Yaonan Jin, Tim Randolph, and Rocco Servedio
27th International Conference on Randomization and Computation
View Online

A Note on the Complexity of Private Simultaneous Messages with Many Parties
Marshall Ball and Tim Randolph ITC 2022
3rd Annual Conference on Information-Theoretic Cryptography
View Online

[†]In theoretical computer science, authors are always listed alphabetically by last name. Equal contribution is assumed unless otherwise stated. Conferences are the primary publication venues; journals may be primary publication venues but are often secondary.

Average-Case Subset Balancing Problems SODA 2022
Xi Chen, Yaonan Jin, Tim Randolph, and Rocco Servedio
33rd Annual ACM-SIAM Symposium on Discrete Algorithms
View Online

Parallel Lotteries: Insights from Alaskan Hunting Permit Allocation
Nick Arnosti and Tim Randolph MS 2021; EC 2021
Management Science Vol. 68, No. 7 (Journal version)
22nd ACM Conference on Economics and Computation, as "Alaskan Hunting License Lotteries are Flexible and Approximately Efficient" (Conference abstract)
View Online

A Lower Bound on Cycle Finding in Sparse Digraphs SODA 2020; TALG 2022
Xi Chen, Tim Randolph, Rocco A. Servedio, and Tim Sun
ACM Transactions on Algorithms, Vol. 18, Issue 4 (Journal special issue)
31st Annual ACM-SIAM Symposium on Discrete Algorithms (Conference)
View Online

(k, p)**-Planarity: A Relaxation of Hybrid Planarity** WALCOM 2019; TCS 2021
Emilio di Giacomo, William J. Lenhart, Giuseppe Liotta, Timothy W. Randolph, and Alessandra Tappini
Theoretical Computer Science, Vol. 896 (Journal special issue)
13th International Conference and Workshops on Algorithms and Computation (Conference)
View Online

Tight Bounds for $(t, 2)$**-Broadcast Domination on Finite Grids** RHUMJ 2019
Timothy W. Randolph
Rose-Hulman Undergraduate Mathematics Journal, Vol. 20
View Online

Optimal (t, r)**-Broadcasts on the Infinite Grid** DAM 2019
Benjamin F. Drews, Pamela E. Harris, and Timothy W. Randolph
Discrete Applied Mathematics, Vol. 255
View Online

Invited Articles

Surveying Theory of Computation Education in the United States EATCS 2026
Ryan E. Dougherty and Tim Randolph
Bulletin of the European Association for Theoretical Computer Science No. 148
View Online

Preprints

Peer-to-Peer Tutorials in Algorithms: Reflections on Scaling Up Interactive Presentation-Based Learning Preprint, 2026
Tim Randolph

Exact Algorithms for Finding Sumsets Preprint, 2023
Tim Randolph
View Online

Invited Talks, Posters, and Panels

Research Presentations

"A Peer-to-Peer Tutorial Model for CS Theory and Algorithms" (talk)
Symposium on Theory of Computing (STOC), Salt Lake City, UT, 6/23/2026

"Pedagogy in Theory of Computing and Algorithms" (panel)
ACM Technical Symposium on Computer Science Education (SIGCSE), St. Louis, MO, 2/20/2026

"Exact and Parameterized Algorithms for Hard Problems" (talk)
Pomona College, Claremont, CA, 11/6/2025

"Subset Balancing: A Puzzle of Algorithmic (Additive?) Combinatorics" (talk)
University of California Irvine, Irvine, CA, 10/17/2025

"Exact and Parameterized Algorithms for Subset Sum Problems" (thesis defense)
Columbia University, New York, NY, 4/9/2024

"Designing Algorithms for Hard Problems: A Case Study" (talk)
Williams College Computer Science Colloquium Series, Williamstown, MA, 4/5/2024

"Experience Report: Participatory Governance in the Computer Science Theory Classroom" (talk), *ACM Technical Symposium on Computer Science Education (SIGCSE)*, Portland, OR, 3/21/2024

"Algorithmic Approaches to Subset Sum (And Other Hard Problems)" (talk)
Harvey Mudd College, Pomona College, Amherst College, Bard College, Union College, 10/31/2023-11/25/2023

"Log Shaving for Subset Sum" (talk)
27th International Conference on Randomization and Computation (RANDOM 2023), Atlanta, GA, 9/12/2023

"The Complexity of PSM with Many Parties" (talk)
3rd Conference on Information-Theoretic Cryptography (ITC 2022), Boston, MA, 7/6/2022

"Average-Case Subset Balancing Problems" (talk)
31st Annual Symposium on Discrete Algorithms (SODA 2022), Virtual, 1/9/2022

"Parallel Lotteries: Insights from Alaskan Hunting Permit Allocation" (poster)
22nd Conference on Economics and Computation (EC 2021), Virtual, 7/21/2021

"Alaskan Hunting License Lotteries are Flexible & Approximately Efficient" (talk, poster)
DSI Financial and Business Analytics Center, New York, NY, 11/12/2019
15th Conference on Web and Internet Economics (WINE 2019), New York, NY, 12/10/2019

"The Case for Wasteful Allocation Mechanisms" (talk, poster)
1st INFORMS Workshop on Market Design, Phoenix, AZ, 6/28/2019
3rd Workshop on Mechanism Design for Social Good (MD4SG 2019), Phoenix, AZ, 6/28/2019

"(k,p)-planar Drawings of Cluster Graphs" (talk)
Williams College Summer Science Expo, Williamstown, MA, 8/11/2017

Outreach Presentations

"Graduate School in Computer Science" (panel)
Harvey Mudd College Summer of CS, Claremont, CA, 7/8/2025
Southern California REU Symposium, Claremont, CA, 7/31/2025

"Navigating Graduate School in (Theoretical) Computer Science" (talk)
Columbia Emerging Scholars Program (ESP) Research Symposium, New York, NY, 4/12/2024

"Research and Exploration in (Theoretical) Computer Science" (talk)
Columbia Engineering Summer High School Academic Program (SHAPE), New York, NY, 8/11/2022

"Demystifying the Dissertation: Research in Theoretical Computer Science" (talk)
Columbia University Demystifying the Dissertation Seminar Series, New York, NY, 12/9/2020

"Research in Algorithms and Mechanism Design" (talk)
Columbia Emerging Scholars Program (ESP) Research Symposium, New York, NY, 11/20/2020

"Demystifying the PhD: Applying to PhD Programs" (talk)
Columbia University PhD Project Presentation Series, New York, NY, 11/18/2020

Grants and Awards

Professional Development Network Grant, Claremont Colleges Faculty Development *Funding to create a community of practice for CS theory at the Claremont Colleges.*	2026
NSF REU Mentor, Harvey Mudd College. *Grant no. 2243941.*	2025
Internal Award for Summer Research, Harvey Mudd College	2025
Michelman Award for Exemplary Service, Columbia University	2022
Columbia CS Department Service Award, Columbia University (3x)	2020, 2021, 2023
Sam Goldberg Prize, Williams College	2018
Elected Member, Sigma Xi	2018
Class of 1960s Scholar in Computer Science, Williams College (2x)	2017, 2018
Elected Member, Phi Beta Kappa (Junior Year)	2017
Class of 1960s Scholar in Cognitive Science, Williams College	2017

Service

Professional Service

Conference and Journal Reviews

ACM Conference on International Computing Education Research (ICER)
ACM Symposium on Theory of Computing (STOC)
ACM-SIAM Symposium on Discrete Algorithms (SODA)
ACM Special Interest Group on Computer Science Education Technical Symposium (SIGCSE)
ACM Transactions on Computing Education (TOCE)
Consortium for Computing Science in College: Southwestern Region (CCSC-SW)
European Symposium on Algorithms (ESA)
International Colloquium on Automata, Languages, and Programming (ICALP)
Symposium on Simplicity in Algorithms (SOSA)
Symposium on Theoretical Aspects of Computer Science (STACS)

Session Chair

ACM-SIAM Symposium on Discrete Algorithms (SODA 2022)

Institutional Service

MAP-CS Working Group, Harvey Mudd College 2026-
Long-term curriculum redesign committee participating in the 2026-2027 MAP-CS cohort.

Faculty Budget Committee, Harvey Mudd College 2025-
Authored "HMC Faculty Guide to the Budget".

CS Curricular Redesign Committee Member, Harvey Mudd College 2024-2026
Committee work during the redesign of the HMC introductory CS sequence. Participated in Capacity, External Compatibility, Mathematical Foundations, and Computational Thinking working groups.

PhD Student Representative, Columbia University 2022-2024
PhD Coordinator, Columbia University Emerging Scholars Program 2019-2022
Union Organizer, Student Workers of Columbia (UAW Local 2710) 2021-2022
Founding Organizer, Columbia Pre-Submission Application Review Program 2020-2021
Founding Organizer, Columbia Graduate Student Theory Retreat 2019-2021
Speaker, Columbia "Demystifying the Dissertation" Initiative 2020-2021

Undergraduate Research Students

Summer Research:
Mia Alexander (HMC '26), Theo Julien (HMC '26), Xiaofeng Li (HMC '27), Riley Brown (University of Nebraska-Lincoln '27), Uddant Patodia (HMC '28)

Independent Study:
Uddant Patodia (HMC '28)

Advising and Mentorship

External Capstone Advisor, U.S. Military Academy at West Point 2024-2025
Advised on "TheoryViz: A Visualizer Tool for Theory of Computing Concepts". This capstone appeared as a poster in SIGCSE '25.

Mentor, Williams CS Alumni Mentorship Program 2022-2024
Mentor, Women in Science at Columbia (WISC) 2021-2024

Mentor, Lumiere Research Scholars Program 2022
Mentor, Barnard Better, Enhance, and Advance Research Series (BEARS) 2022
Advisor, Columbia Undergraduate Theory Seminar 2022